

5.2 Distributed Representations

Similar words share dimensions instead of a single one-hot slot.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

1. Why Week 4 is not enough

Basic vectorization (one-hot, BoW, n-grams, TF-IDF) shares three failures:

Failure	What it means in a project
Discrete / atomic	<i>film</i> and <i>movie</i> are unrelated coordinates
Sparse, high-dimensional	Most of the vector is zero; learning and storage suffer
OOV	A new word has no column (fastText is a later exception)

Distributed representations are the response: short, dense vectors whose directions encode meaning. Word2Vec (notes **5.3–5.5**) is the method this course uses first.

Sources: *Practical Natural Language Processing* and Lane, Howard, Hapke, *Natural Language Processing in Action*.

2. Three terms people mix up

Distributional and **distributed** are not the same word.

A **distributional** representation comes from how a word is **used**. You look at the contexts it appears in. The raw object can still be huge: a row of a co-occurrence matrix, or a one-hot. Harris's idea, in slogan form: words that occur in similar contexts mean similar things.

Example corpus: I like deep learning. I like NLP. I enjoy flying. A windowed co-occurrence matrix counts how often two words share a neighborhood:

$$X = \begin{matrix} & \begin{matrix} I & like & enjoy & deep & learning & NLP & flying & . \end{matrix} \\ \begin{matrix} I \\ like \\ enjoy \\ deep \\ learning \\ NLP \\ flying \\ . \end{matrix} & \begin{bmatrix} 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

Each row is still high-dimensional and still mostly about counts.

A **distributed** representation **compresses** that. The vector is low-dimensional (50–300 is typical) and **dense** (almost no zeros). Meaning is spread across the coordinates. No single index “is” the word *dog*.

An **embedding** is the map from the distributional space (or from word ids) into that dense space. “We trained embeddings” means: we learned that map.

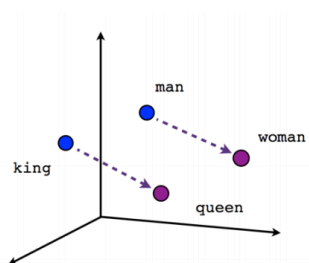
3. What the geometry is for

If the map is good, **distance and direction mean something**.

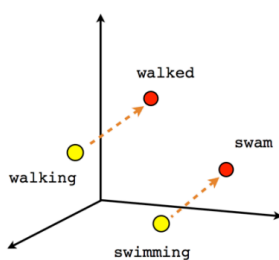
Mikolov et al. showed that Word2Vec vectors support analogies:

$$[\vec{\text{king}} - \vec{\text{man}} + \vec{\text{woman}}] \approx \vec{\text{queen}}$$

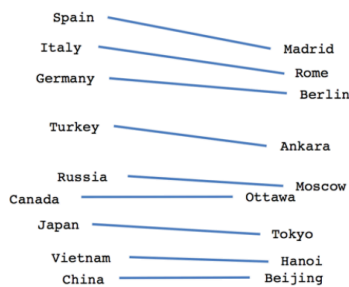
The same kind of offset shows up for gender, verb tense, and country–capital pairs:



Male-Female



Verb tense



Country-Capital

You are not memorizing the picture. You are taking the claim: a useful embedding turns a linguistic relation into a **vector offset** you can add.

Nearest neighbors of a word (cosine) become a debug tool. If *Paris* sits next to *France* and *Berlin*, the space learned something. If *Paris* sits next to random tokens, training failed or the corpus is wrong.

4. How this is learned, in one paragraph

You do **not** build the huge co-occurrence matrix and then factor it (that is a different family: LSA, GloVe). Word2Vec trains a **small two-layer net** on a fake task: predict a word from its neighbors, or the neighbors from a word. After training you **throw away the softmax** and keep the hidden-layer weights. Those rows are the embeddings.

CBOW (note 5.3) predicts the center from the context. Skip-gram (note 5.4) predicts the context from the center. The practical tricks that make either one scale are note 5.5.

5. What you should walk away with

- Distributional = from context. Distributed = dense and compressed.
- Embeddings are a learned map into the dense space.
- Cosine in that space is the default similarity.
- Week 4 vectors cannot do *king* – *man* + *woman*. These can, when training worked.

What you will type. In gensim, `model.wv.most_similar("king")` and `model.wv.most_similar(positive=["king", "woman"], negative=["man"])` are the two checks. If the first list is junk, do not bother with the analogy. Bad neighbors mean a bad corpus, a too-small window, or too few epochs—not a bad formula.

6. Practice

1. In one sentence each, distinguish distributional representation, distributed representation, and embedding.
2. Why is a 300-dimensional Word2Vec vector less prone to the sparsity problem than a 50,000-dimensional TF-IDF vector?
3. You compute $\vec{\text{Paris}} - \vec{\text{France}} + \vec{\text{Italy}}$ and the nearest word is *Rome*. What did the model learn?